CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CE/EE/IT/Backlog) Semester – II Examination, 2011
EE - 101 Fundamentals of Electrical & Electronics Engineering

Date: 12.05.2011, Thursday Time: 01:30p.m to 04:30p.m Maximum Marks: 70

Instructions:
1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary.

SECTION I

Q-1 **A.** Show that the resonant frequency of an *R-L-C* series circuit is given by $f_r = \frac{1}{2\pi\sqrt{LC}}$ and Also derive expressions for *Q*-factor. **4**

B. A sinusoidal voltage has a value of 100 volts at 2.5 ms and it takes time of 30 ms to complete one cycle. Find the maximum value and time to reach it for the first time after zero. **3**

C. A parallel circuit consisting of the following three branches is connected to a 100 V ac supply. **5**
Branch A: purely resistive, resistance, 10 Ω
Branch *B:* inductive, resistance, 4 Ω, reactance, 6.28 Ω
Branch *C:* capacitive, resistance, 8 Ω reactance, 15.91 Ω
Calculate (i) current in each branch, (ii) total current taken from the supply and (iii) power factor of the combined circuit.

OR

Q-1 **A.** Derive an expression for the instantaneous value of alternating sinusoidal emf in terms of its maximum value, angular frequency and time. **6**

B. Drive the relationship between line voltage and phase voltage, line current and phase current in case of 3 phase delta connected system. **3**

C. Prove that the current in a purely capacitive circuit leads its voltage by 90°. **3**

Q-2 **A.** State and explain coulomb's laws for electric circuits. **3**

B. A condenser and a resistor are connected in series with a d.c. source of V volts. Derive an expression for the voltage and current during the charging of capacitor. And also show the volt – time and current – time characteristic. **5**

C. A parallel plate capacitor has plates of area 2 m^2 spaced by the three slabs of different dielectrics. The relative permittivities are 2, 3 and 6 and the thicknesses are 0.4, 0.6 and 1.2 mm respectively. Calculate the combined capacitance and the electric stress in each material when the applied voltage is 1000 V. **3**

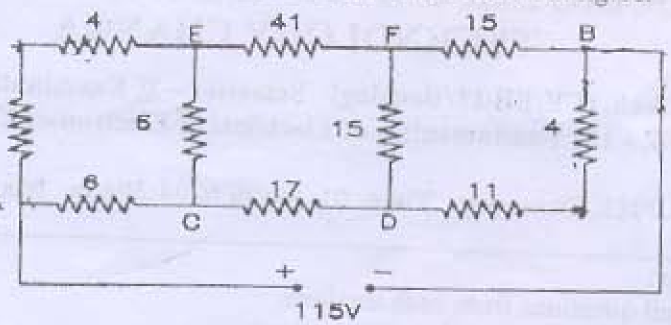
OR

Q-2 **A.** Give the comparison between series and parallel circuits. **4**

B. Explain why domestic appliances are not connected in series. **2**

P.T.O.

C. Determine the current in the 17Ω resistor in the network shown in figure. 5



Q-3 A. Explain hysteresis loss and eddy current loss. 6

B A coil is wound uniformly with 300 turns over a steel ring of relative permeability 900 having a mean diameter of 20 cm. The steel ring is made of a bar having cross-section of diameter 2 cm. If the coil has a resistance of 50 ohms and is connected to 250 V d.c. supply. Calculate (i) the m.m.f. (ii) the field intensity in the ring (iii) reluctance of the magnetic path (iv) total flux and (v) permeance of the ring. 6

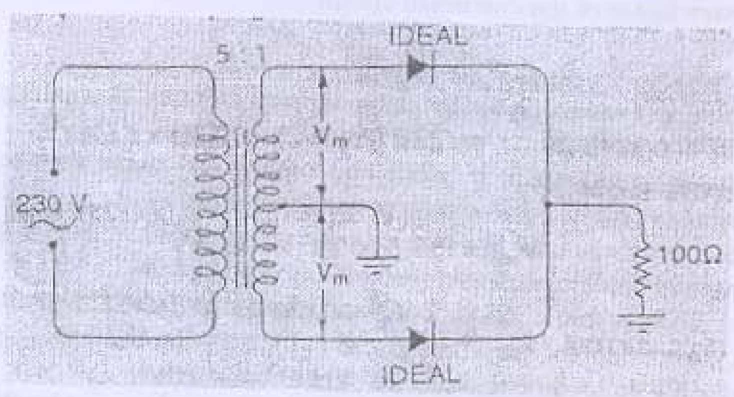
SECTION – II

Q-4 A. Explain the *V-I* characteristics of diode. Also explain the operation of half-wave rectifier. 5

B. Explain the operation of Positive clipper. 3

C. In centre-tap rectifier shown in below figure, the diodes are assumed to be ideal and having zero internal resistance, find 4

(i) DC output voltage, (ii) Peak Inverse Voltage, (iii) Rectification efficiency



Q-5 A. Explain different filters circuits in detail. 7

B. Explain following Logic Gates in detail (i) OR gate, (ii) NAND gate. 2

C. Explain any two of following terms: 2

a. Duty cycle

b. Rise time

c. Periodic signal

[P.T.O.]

OR

Q-5	A.	(I) what is the limitation of digital system over analog system?	7
		(II) (a) Perform $(11110)_2$ - $(10100)_2$ using 1's complement. (b) Perform $(261)_8$ - $(155)_8$	
	B.	Convert given Binary code to Gray code (i)1101010, (ii)1011100	2
	C.	(i) Convert $(100010111)_2$ number into Hexadecimal number. (ii) Convert $(BC89)_{16}$ number into octal number.	2
Q-6	A.	Derive the relationship between α and β.	4
	B.	Give the advantages of transistor over vacuum tube. Also explain construction of the transistor.	4
	C.	Describe the transistor action in detail.	4

OR

Q-6	A.	Name the three possible transistor configurations. Also show that α is always less then unity.	5
	B.	Explain common base configuration in detail and also derive the useful equations.	4
	C.	Explain the characteristics of common emitter transistor.	3